Data Analysis Software: The EasyScience Framework


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Framework

easy science

EasyScience (easyscience.org) is a framework initiated at the European Spallation Source (ess.eu) to help unify and simplify data analysis across neutron scattering techniques.

It aims to make the analysis step more accessible by providing user-friendly graphical interfaces for new users, while still supporting flexible script-based workflows in Jupyter notebooks for more advanced users.

Core Modules

easy science

EasyScience core library - building blocks for libraries and applications

backend

easy application

Graphical user interface components for desktop applications

frontend

Domain-Specific Projects

easy diffraction

Diffraction data analysis desktop application & python library

model refinement

easy imaging

NEW

Bragg-Edge Imaging data analysis desktop application & python library

model refinement

easy shapes

NEW

Molecular structure generation desktop application

multiscale simulation workflows

easy reflectometry

Reflectometry data analysis desktop application & python library

model refinement

easy dynamics

NEW

Quasielastic neutron scattering data analysis desktop application & python library

model refinement

easy texture

External Project

Texture data reduction desktop application

data reduction

Main Features

Free and Open-Source

EasyScience applications and libraries are freely available for use. The full source code is openly shared on GitHub.

Cross-Platform

EasyScience-based applications run on all major operating systems: macOS, Windows, and Linux.

Improved User Experience

The graphical interface has undergone multiple improvements. Reviewed by UX experts from Design Psychology.

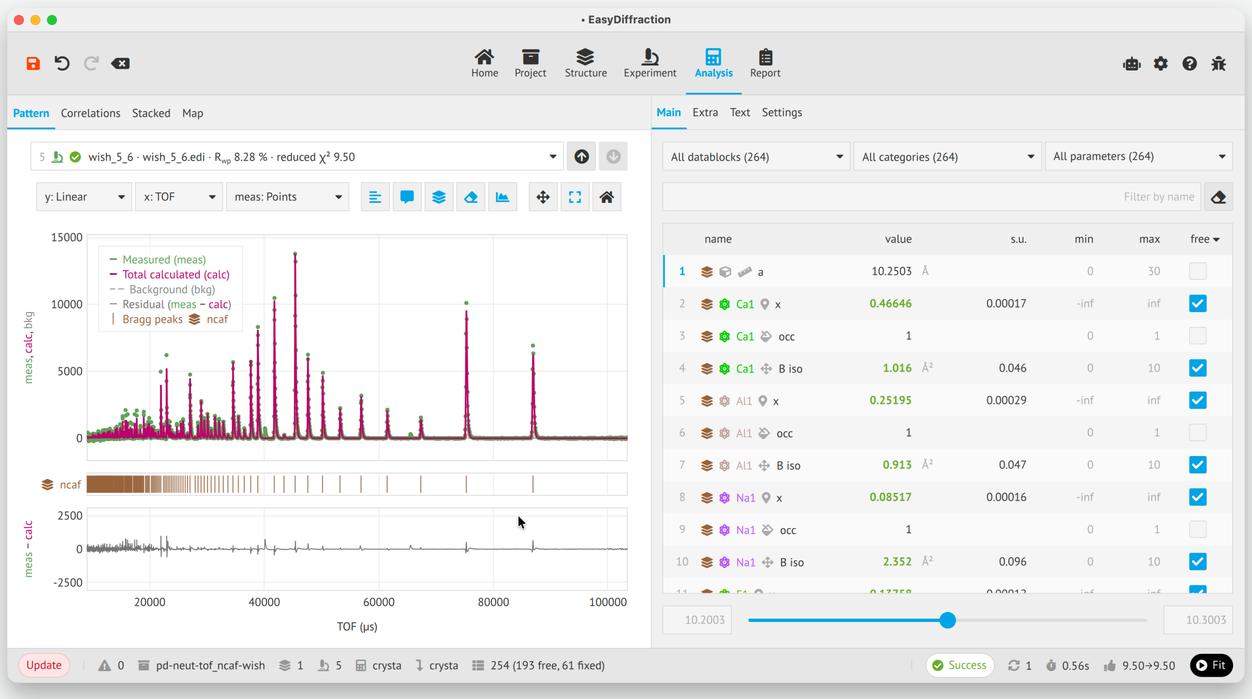
Integrates External Calculators

EasyScience simplifies the integration of domain-specific scientific libraries as external calculation engines.

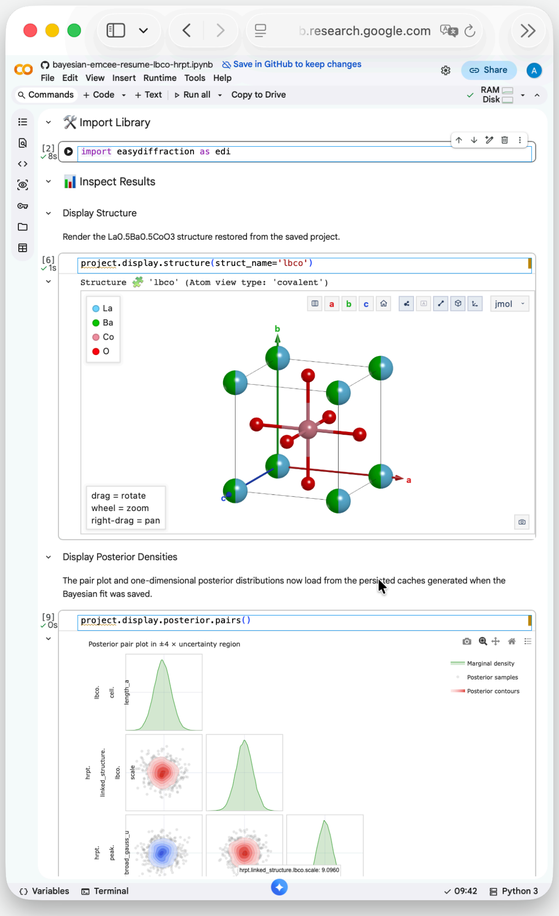
Adaptable to Different User Groups

EasyScience supports users with different data analysis preferences through both graphical and scripting interfaces.

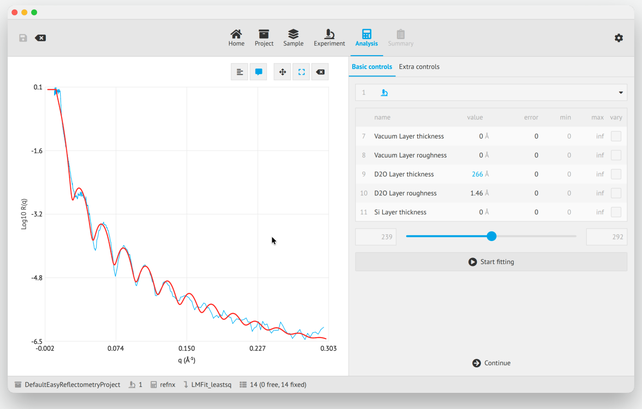
Graphical Interface - EasyDiffraction



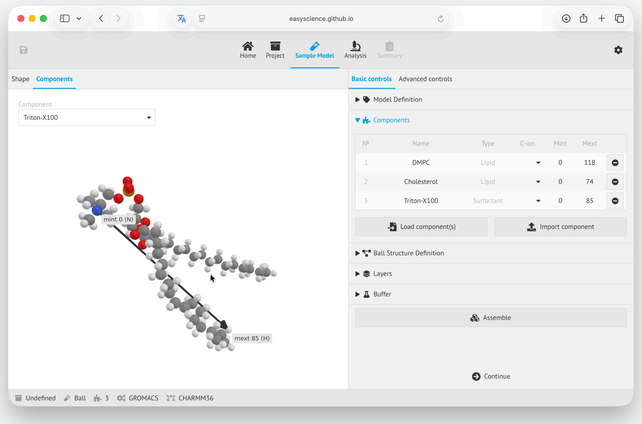
Jupyter Notebook - EasyDiffraction



Graphical Interface - EasyReflectometry



Graphical Interface Prototype - EasyShapes



Graphical App and Python Library

The EasyScience framework supports the development of both graphical desktop applications and Python-based libraries.

Reusable Components

A modular architecture enables reuse of generic components across techniques, improving consistency and maintainability.

Consistent Look and Feel

Shared components ensure a uniform, user-friendly design across domain-specific applications, while allowing focus on scientific features.

Open for Contribution

We are open to collaboration!
Current contributors include
U Göttingen, JCNS @ MLZ, Chalmers U

Feedback

We would love to hear your feedback!
easyscience.org/#contact